Modular Flow of a Gaussian Detector in an Expanding Universe: Finite-N Dynamics and the Breakdown of the Markovian Approximation in FRW

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Abstract

We test a reformulation of the Universal Applied Time (UAT) framework based on the modular flow of Tomita–Takesaki. The original UAT proposal—time as a measure of the causal connection between an event and an observer—is recast as a concrete, computable question about the quantum Fisher information of a Gaussian detector coupled to a scalar field in a Friedmann–Robertson–Walker (FRW) universe. We implement the calculation for a detector with N field modes in a matter-dominated background $a(\eta) = \eta^2$, using symplectic covariance matrices and the standard Gaussian formula for F_Q.

Three results are reported. First, the Unruh–DeWitt consistency check (KMS condition) passes at machine precision ($\max|\Delta W/W| \lesssim 10^{-14}$ for accelerations $a \in \{0.5, 1.0, 2.0\}$), which validates the numerical apparatus. Second, the maximum information mutual $I(O{:}\phi)_{\max}$ between detector and field converges with N to approximately 0.26 nats, and the corresponding informational bound on clock resolution is $\Delta\tau \gtrsim T_{\text{total}}\, e^{-I_{\max}/2}$. The power-law exponent α of the Fisher-information decay does *not* converge with N; we show this is a physical consequence of the finite-dimensional unitary dynamics (quasi-periodic orbits, no genuine decay), not a numerical artefact.

Third, we examine two distinct continuum models of the field. A *static-bath* quantum Brownian motion model with Ohmic spectral density gives an exponential relaxation rate $\Gamma = \pi\eta/2$, independent of the effective temperature and the ultraviolet cutoff, and linear in the coupling. However, an *exact* treatment using the Wightman function of a massless scalar field in FRW shows that the effective damping rate γ_{eff} is not constant: it decays with the expansion due to dilution of the bath, and exhibits sign-changing oscillations (backflow, associated with particle creation) in the matter- and dark-energy-dominated regimes. The Markovian result $\Gamma = \pi\eta/2$ therefore describes a detector coupled to a static external bath, *not* to a cosmological scalar field. The specific numerical values of γ_{eff} depend on the choice of UV regulator and IR subtraction scheme, and are not quoted as physical predictions; the qualitative behaviour is robust.

1 Introduction

The Universal Applied Time (UAT) framework—also referred to by its Spanish acronym TAU, *Tiempo Aplicable Unificado*—was developed independently by the author during 2025 and proposed that the time assigned to an event is a measure of the causal connection between that event and an observer. The original formulation expressed this intuition through additive equations with multiplicative factors encoding cosmological, relativistic, and quantum-gravitational corrections. Systematic tests against DESI DR2 baryon acoustic oscillation data and Pantheon

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supernova data established that the additive formalism was not competitive with ΛCDM, and subsequent analysis identified the root cause as a structural incompatibility between the relationalist intuition and the additive formalism (5).

The present paper does not attempt to salvage the original equations. Instead, it takes the underlying intuition seriously and asks what mathematical object it actually points to. The answer—developed in collaboration with automated reasoning tools and formalised in a companion specification—is that the intuition reduces, in the calculable regime, to a statement about the *informational cost* of maintaining temporal resolution in an expanding universe.

1.1 Relation to the modular program

The technical framework of this paper is the modular theory of Tomita–Takesaki (6; 7), as applied to quantum clocks in cosmological settings by recent work (9; 10; 11; 12; 13; 14; 15).

The UAT/TAU framework was developed independently by the author in 2025, from a starting point in cosmological numerical modeling and direct communication with experimental collaborations. Its public development record is documented in Section 11. For the reader's reference, the following dates are relevant:

- Chen and Penington formalised the modular treatment of cosmological clocks in June 2024 (10).
- De Vuyst and collaborators developed the crossed-product and quantum-reference-frame formulation of gravitational entropy in December 2024 (13).
- Speranza proposed the intrinsic cosmological observer in April 2025 (12).
- The UAT/TAU framework was communicated to the CMB-S4 experimental collaboration in April 2025, and formally submitted to peer-reviewed venues from May 2025 onward (Section 11).
- Chen and collaborators extended the crossed-product approach to covariant observers in de Sitter space in November 2025 (11).

The convergence of UAT with the modular program is therefore independent and parallel. The framework does not claim priority over **(author?)** (10, 13), whose work predates it. What UAT contributes is a phenomenological intuition developed outside the algebraic quantum field theory community, from a starting point in cosmological numerical modeling. The present paper takes that intuition, translates it into the language of modular theory, and provides its first quantitative test.

2 What UAT/TAU Really Points To

Before the technical development, we state the interpretive claim in one sentence, because it is the only claim this paper endorses:

> **The capacity of a quantum observer to resolve cosmic time is governed by its mutual information with the field.**

This is the honest translation of the original UAT/TAU intuition to the language of modular theory and quantum information. It replaces the earlier language of "applied time" and "causal connection" by a quantity that is mathematically well defined, computable, and falsifiable in principle: the quantum Fisher information of the detector's reduced state with respect to the cosmic time parameter.

No stronger claim is made in this paper. In particular, we do not assert that time "runs differently" for different observers, nor that the metric is modified, nor that any cosmological parameter is altered. The observable is the resolution of a clock, not the rate of time.

3 Model

3.1 Background and field

We work in a spatially flat FRW universe with line element

$$ds^2 = -dt^2 + a^2(t)\, d\vec{x}^2, \tag{1}$$

and adopt the matter-dominated regime $a(\eta) = \eta^2$ in conformal time η, which is sufficient for the numerical demonstration. A real massive scalar field ϕ of mass $m_\phi \ll H_0$ is expanded in comoving modes,

$$\phi(\vec{x}, \eta) = \frac{1}{\sqrt{L^3}} \sum_{\vec{k}} \left[a_{\vec{k}} \phi_k(\eta) e^{i\vec{k}\cdot\vec{x}} + \text{h.c.} \right], \tag{2}$$

with the mode functions satisfying the standard FRW equation. The field is prepared in the Bunch–Davies vacuum at early times.

3.2 Detector

The observer is modelled as a harmonic oscillator of frequency ω_0 with canonical quadratures $R_O = (x_O, p_O)$ obeying $[x_O, p_O] = i$ ($\hbar = 1$). We use a harmonic oscillator rather than a two-level system because the bilinear coupling

$$H_{\text{int}} = g\, x_O\, \phi(\vec{x}_O, \eta) \tag{3}$$

preserves Gaussianity of the joint state, reducing the entire problem to matrix algebra on covariance matrices. A two-level detector coupled through σ_x would break Gaussianity and would in addition be subject to the bound $I(O{:}\phi) \leq \ln 2$, which truncates the analysis prematurely.

3.3 Covariance matrix formalism

The joint state is Gaussian and fully characterised by the covariance matrix

$$\sigma_{ij} = \tfrac{1}{2} \langle \{ R_i - \langle R_i \rangle, R_j - \langle R_j \rangle \} \rangle, \tag{4}$$

with $R = (x_O, p_O, x_1, p_1, \ldots, x_N, p_N)$ for N field modes retained. The symplectic eigenvalue $\nu_m \geq 1/2$ of each 2×2 block determines the von Neumann entropy

$$S(\nu) = (\nu + \tfrac{1}{2}) \ln(\nu + \tfrac{1}{2}) - (\nu - \tfrac{1}{2}) \ln(\nu - \tfrac{1}{2}). \tag{5}$$

The information mutual between the detector and the field is

$$I(O{:}\phi) = S(\sigma_O) + S(\sigma_\phi) - S(\sigma_{\text{total}}). \tag{6}$$

3.4 Quantum Fisher information

The Fisher information of the detector state with respect to a time parameter λ is, for a Gaussian state,

$$F_Q^{(\lambda)} = \tfrac{1}{2} (\partial_\lambda \sigma)^T (\sigma \otimes \sigma - \tfrac{1}{4} \Omega \otimes \Omega)^{-1} (\partial_\lambda \sigma), \tag{7}$$

where Ω is the symplectic form and the derivative is taken with respect to λ. The Cramér–Rao bound then gives the resolution limit

$$\Delta\lambda \geq \frac{1}{\sqrt{F_Q^{(\lambda)}}}. \tag{8}$$

Two distinct time parameters are physically relevant:

- the detector's proper time τ_O, and
- the cosmic (conformal or scale-factor) time η or t.

For an expanding universe, $dt = a\,d\eta$, so $F_Q^{(t)} = F_Q^{(\eta)}/a^2$. Part of any apparent degradation of clock resolution is therefore purely kinematic. Any genuine dynamical effect must be identified after removing this factor.

4 Unruh–DeWitt Consistency Check

Before computing in FRW, we verify the apparatus against the canonical result for an accelerated detector in Minkowski space. The Wightman function along a uniformly accelerated trajectory in $1+1$ dimensions is

$$W(\Delta\tau) = -\frac{1}{4\pi}\ln\left[-\frac{4}{a^2}\sinh^2\left(\frac{a\,\Delta\tau}{2}\right) - i\varepsilon\right]. \tag{9}$$

The KMS condition for a thermal state at inverse temperature $\beta = 2\pi/a$ requires

$$W(\Delta\tau - i\beta) = W(-\Delta\tau). \tag{10}$$

We test this identity numerically for $a \in \{0.5, 1.0, 2.0\}$ and $\tau \in [0.01, 5.0]$. The results are reported in Table 1.

a	$T = a/2\pi$	$\max\lvert\Delta W/W\rvert$
0.5	0.0796	1.0×10^{-14}
1.0	0.1592	5.0×10^{-15}
2.0	0.3183	2.6×10^{-15}

Table 1: KMS condition test. The maximum relative deviation is at the level of double-precision round-off. The apparatus is correct.

The KMS test passes at machine precision. This confirms that the detector-field formalism, the Wightman function, and the numerical implementation are correct. Any subsequent result on the FRW background can be trusted as a genuine physical prediction of the model, not as an artefact of the numerical framework.

5 Results in a Finite Truncation

5.1 Setup

We take N field modes with comoving momenta $k_j = 2\pi j/L$, $j = 1, \ldots, N$, on a comoving box of size $L = 2\pi$. The detector frequency is $\omega_0 = 1$, the coupling is $g = 1$, and the field mass is $m_\phi = 0.25$. The conformal time runs from $\eta_i = 2$ to $\eta_f = 7$, corresponding to $a \in [4, 49]$ in the matter-dominated background $a(\eta) = \eta^2$. The joint system is evolved unitarily with the symplectic propagator, from the vacuum state $\sigma(0) = I/2$.

5.2 Information mutual and its convergence

The maximum information mutual between detector and field during the evolution is reported in Table 2.

Two facts emerge:

N	$\nu_{\max}$	$I(O{:}\phi)_{\max}$	α (envelope fit)
5	0.5252	0.236	-0.04
10	0.5269	0.249	$+0.42$
20	0.5277	0.255	-1.02
40	0.5283	0.259	-1.49
80	0.5285	0.261	-0.66

Table 2: Convergence of observables with the number N of retained field modes. The information mutual $I(O{:}\phi)_{\max}$ converges smoothly. The power-law exponent α of the Fisher information does not converge and changes sign.

- $I(O{:}\phi)_{\max}$ converges monotonically to approximately 0.26 nats as N increases. This is a stable, well-defined observable of the model.
- The exponent α fitted to the envelope of $F_Q^{(\eta)}$ in the window $a \in [15, 35]$ does not converge. It changes sign between successive values of N.

5.3 Interpretation of the non-convergence of α

The non-convergence of α is not a numerical artefact. It is a physical consequence of the finite-dimensional, unitary dynamics of the model:

- The joint system has $2(N+1)$ degrees of freedom and its spectrum is discrete.
- Unitary evolution of a finite-dimensional system produces quasi-periodic orbits, not monotonic decay.
- The apparent "decay" of $F_Q^{(\eta)}$ in any finite window is an interference pattern between nearby modes, not a genuine power-law.
- Changing N changes the mode frequencies, and therefore changes the interference pattern. The fitted exponent is not a physical quantity.

The earlier reported value $\alpha \approx 0.72$ for $N = 10$ in a different parameter window was an artefact of the specific mode content of that window. It does not survive a systematic scan of N and window.

This is itself a physical result: *in finite-dimensional unitary systems, the modular clock resolution does not decay as a power law.* Any genuine decay requires either dissipation (which is absent in the closed unitary evolution we simulate) or a continuum limit with a nonzero spectral density. The two continuum models we consider next give qualitatively different answers.

6 A Static-Bath Model: Quantum Brownian Motion

The simplest continuum model is to replace the discrete set of N modes by a fixed Ohmic spectral density. This is *not* derived from the Wightman function of a field in FRW; it is a phenomenological closure, equivalent to coupling the detector to an external bath at fixed temperature T_{eff}.

6.1 Spectral density and master equation

We adopt an Ohmic spectral density with exponential ultraviolet cutoff,

$$J(\omega) = \eta\, \omega\, e^{-\omega/\Lambda}, \tag{11}$$

where η is a dimensionless coupling constant and Λ is the cutoff frequency. In the Markovian regime ($\Lambda \gg \omega_0$ and ω_0 well inside the Ohmic band), the reduced dynamics of the detector is described by the quantum Brownian motion master equation in the high-temperature limit,

$$\frac{d\sigma}{d\eta} = M\sigma + \sigma M^T + N, \tag{12}$$

with drift matrix

$$M = \begin{pmatrix} 0 & 1/m \\ -m\omega_0^2 & -\gamma/m \end{pmatrix}, \tag{13}$$

and quantum noise matrix

$$N = \begin{pmatrix} 0 & 0 \\ 0 & m\gamma\omega_0 \coth\!\left(\frac{\omega_0}{2T_{\rm eff}}\right) \end{pmatrix}. \tag{14}$$

The damping coefficient for an Ohmic bath is

$$\gamma = \frac{\pi\eta}{2}. \tag{15}$$

The use of the quantum noise (14) rather than its classical limit $2m\gamma T$ is essential: it guarantees that the detector relaxes to the correct Bose–Einstein equilibrium $\nu_\infty = \frac{1}{2}\coth(\omega_0/2T_{\rm eff})$ rather than to the classical equipartition value.

6.2 Relaxation rate

The homogeneous part of (12) evolves as $\sigma_{\rm hom}(s) \propto e^{Ms}\sigma(0)e^{M^T s}$, whose eigenvalues are $-\gamma/2 \pm i\omega_d$ with $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$. Because σ is bilinear in the solutions of the underlying linear equations, the covariance decays as

$$\|\sigma(s) - \sigma_\infty\| \sim e^{-\gamma s}, \tag{16}$$

and therefore the symplectic eigenvalue ν relaxes with rate

$$\boxed{\Gamma = \gamma = \frac{\pi\eta}{2}.} \tag{17}$$

The relaxation rate is:

- independent of the effective temperature $T_{\rm eff}$;
- independent of the ultraviolet cutoff Λ for $\Lambda \gg \omega_0$;
- linear in the dimensionless coupling η.

6.3 Numerical verification

We integrate (12) for the parameter set $\omega_0 = 1$, $m = 1$, $\eta = 0.1$, and extract Γ from the exponential decay of $|\nu(\eta) - \nu_\infty|$ in the window $\eta \in [30, 90]$. The results are reported in Table 3.

The extracted rate is $\Gamma = 0.157077$ in all cases, versus the theoretical prediction $\Gamma = 0.157080$. The relative deviation is 2×10^{-5}, consistent with numerical round-off. The exponent $R^2 = 1.0000$ confirms that the decay is a clean exponential.

The model defined by (11)–(14) is mathematically consistent and gives a clean result. The question is whether it describes the field in FRW. We show next that it does not.

$T_{\rm eff}$	$\nu_\infty^{\rm analytic}$	$\nu_\infty^{\rm numeric}$	$\Gamma_{\rm ext}$	R^2
0.3	0.5370	0.5370	0.15708	1.0000
0.5	0.6565	0.6565	0.15708	1.0000
1.0	1.0820	1.0820	0.15708	1.0000
2.0	2.0415	2.0415	0.15708	1.0000
5.0	5.0167	5.0167	0.15708	1.0000
10.0	10.0083	10.0083	0.15708	1.0000

Table 3: Extracted relaxation rate Γ for various effective temperatures. The theoretical prediction is $\Gamma = \pi\eta/2 = 0.157080$ for $\eta = 0.1$. The agreement is exact to four significant figures, with $R^2 = 1.0000$ in all cases. The analytic and numerical values of ν_∞ agree to four decimals, confirming that the quantum noise (14) is correctly implemented.

7 Exact FRW Treatment: Bath Dilution

The correct way to obtain the damping kernel for a Gaussian detector coupled to a scalar field in FRW is to compute it from the Wightman function of the field. The result is qualitatively different from the static-bath model of Section 6.

7.1 Wightman function and damping kernel

For a massless scalar field in a spatially flat FRW universe with $a(\eta) = \eta^n$ and in the Bunch–Davies vacuum, the rescaled mode functions are

$$u_k(\eta) = \sqrt{\frac{\pi|\eta|}{4}}\, H_\nu^{(2)}(k|\eta|), \qquad \nu = \left|n - \tfrac{1}{2}\right|, \tag{18}$$

and the physical field is $\phi_k = u_k/a(\eta)$. The Wightman function is

$$W(\eta,\eta') = \frac{1}{2\pi^2}\,\frac{1}{a(\eta)a(\eta')}\int_0^\infty dk\, k^2\, u_k(\eta)\, u_k^*(\eta'). \tag{19}$$

The damping kernel that enters the non-Markovian equation of motion for the detector is

$$\gamma(\eta,\eta') = 2g^2\, \mathrm{Im}\, W(\eta,\eta')\, \theta(\eta-\eta'). \tag{20}$$

The effective damping rate at time η is

$$\gamma_{\rm eff}(\eta) = \int_0^{\eta-\eta_{\rm min}} d\tau\, \gamma(\eta,\eta-\tau)\cos(\omega_0\tau), \tag{21}$$

with $\eta_{\rm min} \sim 1/\omega_0$. Both integrals require a UV regulator. We use an exponential cutoff $\exp(-\varepsilon k)$ with ε of the order of the detector size.

7.2 Analytic cases

In two cases, closed-form expressions are available and have been used in the numerical implementation:

Radiation ($n = 1$). For $a(\eta) = \eta$, the equation for u_k reduces to $u_k'' + k^2 u_k = 0$ (since $a''/a = 0$). The Wightman function is the conformal one,

$$\mathrm{Im}\, W_{\rm conf}(\eta,\eta') = \frac{1}{4\pi^2 a(\eta)a(\eta')}\,\frac{2\Delta\varepsilon}{(\Delta^2+\varepsilon^2)^2}, \qquad \Delta = \eta-\eta'. \tag{22}$$

This is exact and has no infrared divergence.

de Sitter minimal ($n = -1$, $\eta < 0$). For an expanding de Sitter universe with $\eta < 0$, the massless minimally coupled field has an infrared divergence that requires explicit regularization. The analytic expression for the rescaled mode $u_k = (1 - i/k\eta)e^{-ik\eta}/\sqrt{2k}$ gives

$$\mathrm{Im}\, W_{\mathrm{dS}}^{\mathrm{bare}} = -\frac{1}{4\pi^2}\left[\frac{2\eta\eta'\varepsilon\Delta}{(\Delta^2+\varepsilon^2)^2} + \arctan\left(\frac{\Delta}{\varepsilon}\right) - \frac{\Delta\varepsilon}{\Delta^2+\varepsilon^2}\right]. \tag{23}$$

The bare value is negative for $\Delta > 0$, which corresponds to anti-damping; the sign flips under exchange of $H^{(1)} \leftrightarrow H^{(2)}$, but the magnitude is unchanged. The arctan term dominates at late times and is the infrared remnant.

Matter ($n = 2$). For $a(\eta) = \eta^2$, $a''/a = 2/\eta^2$, the mode functions are $u_k = \sqrt{\pi\eta/4}\, H^{(2)}_{3/2}(k\eta)$, and the k-integral must be computed numerically with the regulator (20). The numerical implementation is described in the companion script `frw_wightman.py`.

7.3 Results

We have computed $\gamma_{\mathrm{eff}}(\eta)$ for the three backgrounds. The specific numerical values of γ_{eff} depend on the UV regulator ε, on the infrared subtraction scheme, and on the numerical integration parameters. They are therefore *not* quoted here as physical predictions. What is robust across all choices is the qualitative behaviour, summarized in Table 4.

Background	n	γ_{eff} behaviour	Notes
Radiation	1	Positive, decays with η	Exact conformal case; consistent with bath dilution
Matter	2	Oscillatory, changes sign	Backflow from particle creation; non-Markovian memory
de Sitter	-1	Oscillatory, η-dependent	IR sensitive to subtraction scheme; no constant limit

Table 4: Qualitative behaviour of the effective damping rate from the exact FRW Wightman function. In no case does γ_{eff} approach the static-bath prediction $\pi\eta/2$. The specific numerical values depend on the UV regulator ε and on the IR subtraction scheme, and are not quoted as physical predictions. The qualitative behaviour is robust across choices.

Three features are robust across choices of regulator and numerical implementation:

1. γ_{eff} decreases with η (or with a) rather than approaching a constant. The static-bath prediction $\Gamma = \pi g^2/2$ is not reproduced.

2. In the matter regime, γ_{eff} changes sign as a function of η, indicating backflow: the detector returns information to the field via particle creation. A Markovian dissipation term, which is by construction positive, cannot reproduce this behaviour.

3. In de Sitter, the massless minimally coupled field has an infrared divergence that is regularized differently depending on the subtraction scheme. The resulting γ_{eff} is scheme-dependent, but in all schemes it decays as $\eta \to 0^-$ rather than approaching a constant.

7.4 Physical interpretation

The discrepancy between Section 6 and Section 7 has a clear physical origin. In the static-bath model, the bath is an external environment at fixed temperature: modes of all frequencies remain

available, and the detector reaches a thermal equilibrium. In FRW, the bath *is* the scalar field, and its properties change with the expansion:

- **Dilution.** The energy density of the field decays as a^{-4} (radiation) or a^{-3} (matter). The effective coupling between the detector and the field therefore decays with a.
- **Redshifting.** The proper frequency of each comoving mode evolves as $\omega_k(\eta) = k/a(\eta)$. Modes that are resonant with the detector at one epoch are off-resonant at another.
- **Particle creation.** The time dependence of $a(\eta)$ excites the field out of its vacuum, producing particles that back-react on the detector. This is the origin of the negative γ_{eff}.
- **Horizon cutoff.** Modes with wavelength larger than the Hubble radius are frozen and do not contribute to the local damping.

None of these effects is captured by the Ohmic bath of Section 6. The result $\Gamma = \pi\eta/2$ is a mathematically correct property of that model, but the model does not describe a cosmological scalar field.

8 Comparison and Cosmological Implications

The two continuum models give qualitatively different answers:

- **Static bath**: exponential relaxation with $\Gamma = \pi\eta/2$, independent of $a(\eta)$.
- **FRW field**: suppression with the expansion, plus oscillatory backflow, plus anti-damping in de Sitter.

Only the second corresponds to a scalar field in an expanding universe. Its implications are:

8.1 Temporal resolution in FRW

In the FRW case, because γ_{eff} decays, the detector does not lose coherence at a constant rate. Instead, it retains coherence for longer as the universe expands. The bound on temporal resolution is therefore *not* of the form $\Delta\tau \gtrsim (2/\pi\eta)(1+z)^{-1}$ obtained from the static-bath model. It weakens with time, becoming trivial at late times.

This is a weaker constraint than the static-bath result, but it is the physically correct one for a cosmological field.

8.2 Reinterpretation of the UAT intuition

The original UAT intuition—that the observer's temporal resolution is bounded by its informational connection to the field—survives the correction. What changes is the specific form of the bound: in FRW, the connection weakens with the expansion, so the bound weakens too. At early times ($z \gtrsim 1$), the detector is strongly coupled to the field; at late times, it is effectively decoupled.

This is consistent with the qualitative intuition that "the observer's clock is tied to the universe", but the quantitative form is different from what the static-bath model suggested.

9 What Is Not Claimed

For clarity, this paper does not claim:

1. That time runs at different rates for different observers. The observable is clock resolution, not clock rate.
2. That the metric is modified, or that any cosmological parameter is altered. The background is standard ΛCDM.
3. That the additive formalism of the original UAT is recoverable. It is not. The correct formalism is the modular one.
4. That a power-law decay of clock resolution exists in finite-dimensional unitary systems. It does not.
5. That the relaxation rate $\Gamma = \pi\eta/2$ of Section 6 describes a detector coupled to a cosmological field. It does not. It describes a detector coupled to a static external bath.
6. That the effective damping rate in FRW is constant. It is not; it decays with the expansion and exhibits oscillatory corrections. The exact form of the decay depends on the regularization scheme.
7. That UAT precedes or anticipates the modular program of **(author?)** (10, 13). It does not. The convergence is independent and parallel; see Section 1.1.

10 Future Work

The extensions of this work are:

1. **Full non-Markovian treatment with analytic regulators.** The numerical computation of $\gamma_{\text{eff}}(\eta)$ in Section 7 uses a sharp UV cutoff ε of order the detector size. An analytic treatment with the correct distributional prescription would clarify the renormalization of the detector frequency and isolate the physical damping from the Lamb shift.
2. **Massive field and conformal coupling.** For $m_\phi \neq 0$ or nonminimal coupling $\xi \neq 1/6$, the mode functions have a different IR behaviour. In particular, the Gibbons–Hawking temperature emerges for $m_\phi \sim H$, and the comparison with a thermal bath becomes meaningful.
3. **Full ΛCDM background with era transitions.** The present FRW calculation uses power-law $a(\eta)$. The full ΛCDM case with radiation–matter and matter–dark-energy transitions would show whether the era boundaries leave a signature in γ_{eff}.
4. **Relation to the crossed-product algebra and quantum-reference-frame formulations** (9; 11; 13). The present calculation treats the detector as a classical external system; a fully algebraic treatment would couple the detector's internal Hilbert space to the cosmological modular flow.

11 Conclusions

We have reformulated the original UAT/TAU intuition—time as a measure of causal connection between an event and an observer—as a computable problem in modular theory and quantum information. The reformulation yields:

1. **KMS verification.** The Unruh–DeWitt detector in Minkowski space satisfies the KMS condition at machine precision ($\max|\Delta W/W| \lesssim 10^{-14}$). The apparatus is correct.

2. **Information mutual.** The maximum information mutual between detector and field converges with N to $I_{\max} \approx 0.26$ nats, and the associated informational bound on clock resolution is $\Delta\tau \gtrsim T_{\text{total}}\, e^{-I_{\max}/2}$.

3. **No power-law decay in finite N.** The exponent α of the Fisher-information decay does not converge with N; the earlier value $\alpha \approx 0.72$ was an artefact of the specific mode content.

4. **The Markovian approximation fails in FRW.** A static Ohmic bath gives $\Gamma = \pi\eta/2$, but the exact Wightman function of a scalar field in FRW gives an effective damping rate γ_{eff} that is not constant, does not approach $\pi\eta/2$, and exhibits sign-changing oscillations (backflow) in the matter and dark-energy regimes. The static-bath result is not applicable to a cosmological field.

The original UAT formulation was not competitive with ΛCDM on cosmological observables. The reformulated framework does not claim to be. It claims something more modest and more defensible: the intuition behind UAT is correct in a specific, calculable sense, and that sense is the informational bound on clock resolution. The bound in FRW is weaker than the static-bath result, but it is the physically correct one.

The comparison between the two continuum models is itself a methodological result: *phenomenological closures of the continuum limit must be derived from the exact Wightman function of the field, not assumed.* The Ohmic bath gives the right answer for the wrong system.

Data and Code Availability

All numerical results in this paper are reproducible from the Python scripts `etapaB_v2.py` (finite-N), `etapaC_v5.py` (static-bath QBM), and `frw_wightman.py` (exact FRW Wightman function). The scripts use only `numpy`, `scipy`, and `matplotlib`. They generate the KMS verification, the finite-N convergence table, and the comparison of the two continuum models. Two independent implementations of the FRW calculation were produced and agree in their qualitative behaviour; the implementation included here uses closed-form expressions for the Wightman function where possible.

Public development record

The UAT/TAU framework was developed independently by the author during 2025. For the reader's reference, the following public records document its development. All timestamps are from the originating server and are not reconstructed.

- **April 2025.** Communication to the CMB-S4 experimental collaboration proposing a framework for modeling PBH-induced CMB anisotropies. The concept is referred to in the correspondence as "Observable Time", subsequently renamed "Applied Time".

- **26 May 2025.** Communication to the Institute of High Energy Physics, Chinese Academy of Sciences, on the applied-time framework for cosmology and singularities.

- **30 May 2025.** Submission of the manuscript *A Unified Applicable Time Framework for Modeling Primordial Black Holes and CMB Anisotropies* to INDTEC, C.A.

- **5 June 2025.** Two submission tickets registered at CONICET: `#2025060510000138` and `#2025060510000147`.

- **11 June 2025.** Submission to *Open Astronomy* (De Gruyter) and to SCIELO-CAICYT.
- **12 June 2025.** Submission to *Revista Mexicana de Astronomía y Astrofísica* (`rmga@astro.unam.mx`).
- **October 2025.** Extended version of the original manuscript, *Evolution of Primordial Black Holes and Their Impact on the Cosmic Microwave Background: A Comprehensive Numerical Study Using the Applicable Time Framework*. Contains the additive formalism discussed in Section 1.1; mentioned for completeness, not cited as a physics reference.
- **2025.** Zenodo archives, DOIs `10.5281/zenodo.17411019`, `10.5281/zenodo.17411163`, `10.5281/zenodo.17422770`, `10.5281/zenodo.17695218`.

These records are provided for completeness. They document an independent development path and are not intended as a priority claim with respect to the modular program. The reader is invited to consult them alongside the references in Section 1.1.

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